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Machine Learning & Communication Skills – Sentence Outline

FIELD: How Much Does the Field of Study Shape University GPA Rankings?

Team:	Adam Bajer; Pablo Torres; Ignacio Gimeno
Course:	Machine Learning & Communication Skills
Academic year:	2026–2027
Submission deadline:	15 November 2026, 23:59 Madrid time
Text limit:	2 pages

FIELD: Field Influence on Evaluation Levels across Degrees.

1 Introduction

1. Using one GPA ranking to allocate places across degrees assumes that grades are comparable across fields of study.
   a. Previous research documents differences in grading patterns between departments within the same university [1, 2].
   b. Evidence from 33 US public institutions also shows recurring differences in average course grades between academic disciplines [3].
2. This project examines how field-related grade differences affect pooled GPA rankings and how ranking students within their degree changes who is selected.
   a. We measure differences in observed grades across fields without interpreting them as direct measures of degree difficulty or student ability.
   b. Using separate Spanish, Portuguese and US datasets, we examine field effects and their persistence before assessing whether alternative ranking rules reduce unequal access to selection.

2 Data

3. [Pablo: three complementary sources answer different parts of the question, and they are analysed separately because they share no key.]
   a. [Pablo: Spanish SIIU table: 12,125 university-field-year means, 89 universities, 29 fields, 2015-16 to 2023-24, with quartiles [4]; limits: aggregated, graduates only.]
   b. [Pablo: Portuguese student records: 3,703 students in 15 degrees with standardised entry grades [5].]
   c. The US dataset provides an external comparison using course or section grades from 33 public institutions, whose departments are mapped through institution-specific classifications and name rules to ten broad areas covering the 29 Spanish fields, with enrolment-weighted analysis possible for 29 institutions [3].

3 Methodology

4. [Pablo: the analysis separates shared field effects from university-specific ones and tests them on data the models have not seen.]
   a. [Pablo: variance decomposition with fixed effects, re-estimated with a mixed-effects model with random university and university-by-field effects.]
   b. [Pablo: temporal validation (train 2015–2021, test 2022–2023) with bootstrap intervals over universities.]
   c. [Pablo: confounders (standardised admission grade, size, dropout) and why dropout is treated separately.]

5. [Pablo/Nacho: the selection simulation and the comparison of ranking rules translate grade gaps into places won or lost.]

   a. [Pablo: the simulation approximates within-field grades from mean and quartiles, and its normal assumption was checked against reported medians and maxima.]

   b. Each of the four ranking rules (raw GPA, regression-adjusted GPA, within-degree z-score and within-degree percentile) ranks Portuguese students by their first-semester grade, selects the top 20%, and is evaluated on the same students' second-semester grades, which the rule has not seen.

4 Results and Discussion

6. The field of study explains most of the differences in grades within each university, and each university adds its own stable pattern on top.

   a. With fixed effects, 65% of the within-university variation is a field effect shared by all universities, 28% is specific to each university and 7% is noise, and the mixed-effects model gives almost the same split (61, 29 and 11%).

   b. The university-specific part stays fairly constant over time, with a correlation of 0.84 between consecutive years.

   c. The model that uses each university's past profile best predicts the years it has not seen, with an error of 0.137 against 0.226 when only the shared field effect is used.

7. The differences between fields are not explained by the entry level of their students.

   a. Controlling for admission grade and size barely changes the field effects, whose values before and after correlate at 0.99, and in the Portuguese records the gap between the highest- and lowest-graded degree only falls from 1.51 to 1.30 points.

   b. Dropout explains part of the differences, but it may also be a consequence of stricter grading rather than its cause.

8. The US analysis partly reproduces the Spanish pattern, with a positive association between broad-area grade effects after controlling for university and year.

   a. Across ten comparable areas, Pearson's correlation between the estimated effects is 0.613 and Spearman's rank correlation is 0.612, with eight areas sharing the same effect sign.

   b. Humanities and languages have a positive effect in Spain but a negative effect in the US, while differences between US course grades and Spanish graduates' final averages limit the comparability of the results.

9. Ranking all students of a university on a single raw-GPA list gives very unequal access by field, and the result does not depend on the assumed grade distribution.

   a. In the median university-year, 41% of Nursing graduates reach the top 20%, compared with 6% of Business Administration graduates (interquartile ranges 29–54% and 2–13%).

   b. [Pablo: replacing the normal with a truncated normal changes shares by 0.04 percentage points on average and no field changes rank.]

10. Ranking students within their own degree would make access to a top-20% selection much more equal across degrees, with almost no change in the later performance of the selected students.

    a. The selected students obtain 14.21 out of 20 in the second semester under raw GPA and 14.08 under within-degree percentiles, but the 95% confidence intervals overlap, so the difference is too small to be conclusive.

    b. Under raw GPA some degrees obtain many more places than others, whereas with within-degree percentiles almost all degrees obtain a similar share, as the spread of selection rates across degrees falls from 0.090 to 0.006.

    c. The higher rank stability of raw GPA (0.66 against 0.62) does not make it a better rule, because it partly comes from the gap between degrees repeating every semester, which shows that the inequality is persistent rather than occasional.

11. The data are aggregated means, they include only students who graduate, and admission grades are published only for public universities, so the results describe associations between fields and grades rather than their causes.

5 Conclusions and Future Work

12. Rankings that pool all degrees by raw GPA sort students largely by their field of study, and ranking within each degree corrects this at almost no cost.

13. Comparing students who take the same courses in different degrees would directly measure whether some fields grade more strictly than others.

References

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Declaration of Contributions and AI Use

[All: one entry per member describing what each actually did (analysis, code, sentences, reviews), and every use of AI tools with tool, purpose, output used and verification.]